Wherever you see **red characters**, these need to be replaced by your information. This includes the **< >** symbols!

Loading in Packages

```
library(tidyverse)
```

Loads a package into the R workspace, so you can use the functions and data it contains

Reading in Data

.csv files:

```
dat <- read_csv("data/<NAME OF DATASET>.csv")
```

.txt files:

```
dat <- read_delim("data/<NAME OF DATASET>.txt")
```

Note: The name of the dataset will change.

Assignment Arrow

```
addition <- 1 + 1
```

Assigns a value to the name of a variable

Preview a Dataset

```
glimpse(<NAME OF DATASET>)
```

Note: This is a great way to see the dimensions of the data (rows and columns) and the data types of each variable!

Filtering a Dataset

```
filter(<NAME OF DATASET>,
       <LOGICAL STATEMENT 1>,
       <LOGICAL STATEMENT 2>,
       …
         )

large_adelie_2008 <- filter(penguins,
                            species == "Adelie",
                            body_mass_g > 3000,
                            year == 2008)
```

Filters observations (rows) out of / into a dataframe, where the inputs (arguments) are the conditions to be satisfied in the data that are kept

Note: It makes your code more readable if you put each filter on a new line (hit enter after each comma)!

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Adding Variables (Mutate)

```
mutate(<NAME OF DATASET>,
       <NAME OF VARIABLE> = <FUNCTION OF OLD VARIABLE>
         )

penguins_large <- mutate(penguins,
                         body_mass_kg = body_mass_g / 1000)
```

Creates new variables or modifies existing variables

Calculating Summary Statistics for Numerical Variables

```
summarize(<NAME OF DATASET>,
          <NAME OF STAT> = <STAT FUNCTION>(<NAME OF VARIABLE>)
         )
```

For example, to calculate the mean and median of the `body_mass_g` variable from the `penguins` dataset we have:

```
summarize(nycflights,
          mean_weight = mean(body_mass_g),
          median_mass = median(body_mass_g))
```

The Pipe Operator

```
penguins |>
     filter(species == "Adelie") |>
     summarize(mean_weight = mean(body_mass_g))
```

Sequences of data steps together, where each step is connected with “and then”.

Summary Statistics for Numeric Variables and a Categorical Grouping Variable

```
<NAME OF DATASET> |>
     group_by(<NAME OF GROUPING VARIABLE>) |>
      summarize(<NAME OF STAT> = <STAT FUNCTION>(<NAME OF VARIABLE>)
```

For example, to calculate the mean of the `body_mass_g` variable from the `penguins` dataset BY the `species` variable we have:

```
penguins |>
     group_by(species) |>
     summarize(mean_weight = mean(body_mass_g))
```

Wherever you see **red characters**, these need to be replaced by your information. This includes the **< >** symbols!

Histogram

```
ggplot(data = <NAME OF DATASET>,
       aes(x = <NAME OF VARIABLE>)) +
  geom_histogram(binwidth = <WIDTH OF BINS>) +
  labs(x = "<TITLE FOR THE X-AXIS>")
```

Note: A histogram **must** have the variable on the x-axis!

Boxplot

```
ggplot(data = <NAME OF DATASET>,
       aes(x = <NAME OF VARIABLE>)) +
  geom_boxplot() +
  labs(x = "<TITLE FOR THE X-AXIS>")
```

Note: This boxplot is horizontal. If you want for your boxplot to be vertical, you use **`y =`** instead of **`x =`** . Keep in mind you will need to change the location of you axis label, too!

Side-by-Sode Boxplot

```
ggplot(data = <NAME OF DATASET>,
       aes(y = <NAME OF CATEGORICAL GROUPING VARIABLE>,
           x = <NAME OF QUANTITATIVE VARIABLE>)) +
  geom_boxplot() +
  labs(x = "<TITLE FOR THE X-AXIS>”,
       y = “<TITLE FOR THE Y-AXIS>”)
```

Stacked Bar Plot

```
ggplot(data = <NAME OF DATASET>,
       aes(x = <NAME OF X-VARIABLE>,
           fill = <NAME OF FILL-VARIABLE>)
       ) +
  geom_bar(position = “fill”) +
  labs(x = “<TITLE FOR THE X-AXIS>”,
       fill = “<TITLE FOR THE LEGEND”,
       y = “Conditional Proportion”)
```